

Question 1: Print Numbers from 1 to N

Code:

```
import java.util.Scanner;

public class PrintNumbers{

    public static void main(String args[]){

        int n;

        Scanner sc=new Scanner(System.in);

        System.out.println("Enter a input: ");

        n=sc.nextInt();

        for (int i=1;i<=n;i++){

            System.out.println(i);

        }

    }

}
```

Question 2: Print Multiples of 3 between 1 and N

Code:

```
import java.util.Scanner;

public class Multiple{

    public static void main(String args[]){

        int n;

        Scanner sc= new Scanner(System.in);

        System.out.println("Enter a number");

        n=sc.nextInt();

        for (int i=1;i<=n;i++){

            if(i%3==0){

                System.out.println(i);

            }

        }

    }

}
```

```

        }
    }

}

```

Question 3: Calculate the Factorial of a Number

Code:

```

import java.util.Scanner;

public class Factorial{

    static int calculateFactorial(int n){

        int factorial=1;

        for (int i=1;i<=n;i++){

            factorial=factorial*i;

        }

        return factorial;

    }

    public static void main(String args[]){

        Scanner sc = new Scanner(System.in);

        int n;

        System.out.println("Enter a number");

        n=sc.nextInt();

        int result=calculateFactorial(n);

        System.out.println("Factorial of "+n+" is "+result);

    }

}

```

Question 4: Print Even Numbers from 1 to N

Code:

```
import java.util.Scanner;

public class EvenNumbers{

    public static void main(String args[]){

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a number: ");

        int n=sc.nextInt();

        for(int i=1;i<=n;i++){

            if(i%2==0){

                System.out.println(i);

            }

        }

    }

}
```

Question 5: Sum of Odd Numbers between 1 and N

Code:

```
import java.util.Scanner;

public class OddNumbers{

    public static void main(String args[]){

        int sum=0;

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a number: ");

        int n=sc.nextInt();

        for(int i=1;i<=n;i++){
```

```

        if(i%2!=0){
            sum=sum+i;
        }
    }
    System.out.println("The sum of odd numbers from 1 to "+n+" is: "+sum);
}
}

```

Question 6: Print All Elements of an Array

Code:

```

import java.util.Scanner;

public class ElementsOfArray{
    public static void main(String args[]){
        int[] arr = new int[5];
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter 5 Integers");
        for(int i=0;i<5;i++){
            arr[i]=sc.nextInt();
        }
        for (int i=0;i<arr.length;i++){
            System.out.println(arr[i]);
        }
    }
}

```

Question 7: Find the Sum of All Elements in an Array

Code:

```
import java.util.Scanner;

public class SumOfAllElements{

    public static void main(String args[]){

        Scanner sc = new Scanner(System.in);

        int sum=0;

        int[] arr=new int[5];

        System.out.println("Enter 5 integers: ");

        for (int i=0;i<arr.length;i++){

            arr[i]=sc.nextInt();

        }

        for(int i=0; i<arr.length;i++){

            sum= sum+arr[i];

        }

        System.out.println("The sum of all numbers is: "+sum);

    }

}
```

Question 8: Print All Names in a String Array

Code:

```
import java.util.Scanner;

public class AllNamesInString{

    public static void main(String args[]){

        Scanner sc = new Scanner(System.in);

        String[] arr=new String[4];

    }

}
```

```

        System.out.println("Enter 4 names: ");
        for (int i=0;i<arr.length;i++){
            arr[i]=sc.nextLine();
        }
        for(int i=0; i<arr.length;i++){
            System.out.println(arr[i]);
        }
    }
}

```

Question 9: Find the Largest Element in an Array

Code:

```

import java.util.Scanner;
import java.util.Arrays;

public class LargestElement{

    public static void main(String args[]){
        int[] arr = new int[5];

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter 5 integers: ");
        for(int i=0;i<arr.length;i++){
            arr[i]=sc.nextInt();
        }

        Arrays.sort(arr);

        int largest = arr[arr.length-1];
    }
}

```

```
        System.out.println("The largest element is: "+largest);

    }

}
```

Question 10: Find the Average of Elements in an Array

Code:

```
import java.util.Scanner;

public class AverageOfAllElements{

    public static void main(String args[]){

        float sum=0;

        Scanner sc = new Scanner(System.in);

        int[] arr=new int[5];

        System.out.println("Enter 5 integers: ");

        for (int i=0;i<arr.length;i++){

            arr[i]=sc.nextInt();

        }

        for(int i=0; i<arr.length;i++){

            sum= sum+arr[i];

        }

        float average=sum/5;

        System.out.println("The average of the numbers is: "+average);

    }

}
```

Question 11: Count Positive and Negative Numbers in an Array

Code:

```
import java.util.Scanner;

public class PositiveAndNegative{

    public static void main(String args[]){

        Scanner sc=new Scanner(System.in);

        int[] arr = new int[6];

        System.out.println("Enter 6 integers: ");

        for(int i=0;i<arr.length;i++){

            arr[i]=sc.nextInt();

        }

        int positive=0;

        int negative=0;

        for (int i=0;i<arr.length;i++){

            if(arr[i]>0){

                positive++;

            }else if(arr[i]<0){

                negative++;

            }

        }

        System.out.println("Positive numbers: "+positive);

        System.out.println("Negative number: "+negative);

    }

}
```

Question 12: Sort an Array in Ascending Order**Code:**


```

import java.util.Scanner;

import java.util.Arrays;

public class AscendingOrder{

    public static void main(String args[]){

        int[] arr = new int[5];

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter 5 integers: ");

        for(int i=0;i<arr.length;i++){

            arr[i]=sc.nextInt();

        }

        Arrays.sort(arr);

        for(int i=0;i<arr.length;i++){

            System.out.print(arr[i]+" ");

        }

    }

}

```

Question 13: Check if an Array Contains a Specific Element

Code:

```

import java.util.Scanner;

import java.util.Arrays;

public class SearchElement {

    public static void main(String[] args) {

```

```
Scanner sc = new Scanner(System.in);
```

```
Integer[] arr = new Integer[5];
```

```
System.out.println("Enter 5 integers:");
```

```
for (int i = 0; i < arr.length; i++) {  
    arr[i] = sc.nextInt();  
}
```

```
System.out.print("Enter the number to search: ");
```

```
int number = sc.nextInt();
```

```
if (Arrays.asList(arr).contains(number)) {  
    System.out.println("Found");  
} else {  
    System.out.println("Not Found");  
}  
}
```

Question 14: Find the Index of an Element in an Array

Code:

```
import java.util.Scanner;
```

```
import java.util.Arrays;
```

```
public class FindIndex {
```

```
public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int[] arr = new int[5];

    System.out.println("Enter 5 integers:");

    // Taking input
    for (int i = 0; i < arr.length; i++) {
        arr[i] = sc.nextInt();
    }

    System.out.print("Enter the number to search: ");
    int number = sc.nextInt();

    // Sort the array
    Arrays.sort(arr);

    // Search for the number
    int index = Arrays.binarySearch(arr, number);

    if (index >= 0) {
        System.out.println("The number " + number +
            " is found at index " + index);
    }
}
```

```
    } else {  
        System.out.println("Not Found");  
    }  
}  
}
```

Question 15:Write a program to print the following pattern:

Code:

```
public class Pattern {  
  
    public static void main(String[] args) {  
  
        // Increasing part  
        for (int i = 1; i <= 5; i++) {  
  
            for (int j = 1; j <= i; j++) {  
                System.out.print(i);  
  
                if (j < i) {  
                    System.out.print("*");  
                }  
            }  
  
            System.out.println();  
        }  
    }  
}
```

```

// Decreasing part
for (int i = 4; i >= 2; i--) {

    for (int j = 1; j <= i; j++) {

        System.out.print(i);

        if (j < i) {

            System.out.print("*");

        }

    }

    System.out.println();

}

}

}

```

Question 16: Write a program to print the following pattern:

Code:

```

public class RightAngleTriangle {

    public static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {

            for (int j = 1; j <= i; j++) {

```

```

        System.out.print(j);

        if (j < i) {
            System.out.print("*");
        }
    }

    System.out.println();
}
}
}

```

Question 17: Write a program to print the following pattern:

Code:

```

public class RightAngleOddTriangle {

    public static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {

            for (int j = 1; j <= i; j++) {

                System.out.print(2*j-1);

                if (j < i) {
                    System.out.print("*");
                }
            }
        }
    }
}

```

```

        }
    }

    System.out.println();
}
}
}

```

Question 18: Write a program to print the following pattern:

Code:

```

public class Square{
    public static void main(String args[]){
        for(int i=1;i<=5;i++){
            for(int j=1;j<=5;j++){
                System.out.print(i);
            }
            System.out.println();
        }
    }
}

```

Question 19: Write a program to print the following pattern:

Code:

```

public class Triangle {

    public static void main(String[] args) {

```

```
for (int i = 1; i <= 5; i++) {  
  
    for (int j = 1; j <= i; j++) {  
        System.out.print(i);  
    }  
  
    System.out.println();  
}  
}
```

Question 20: Write a program to print the following pattern:

Code:

```
public class PatternNumbers {  
  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 5; i++) {  
  
            for (int j = 1; j <= i; j++) {  
                System.out.print(j);  
            }  
  
            System.out.println();  
        }  
    }  
}
```



```
}  
}
```

Question 21: Write a program to print the following pattern:

Code:

```
public class NumberTriangle {  
  
    public static void main(String[] args) {  
  
        int number = 1;  
  
        for (int i = 1; i <= 5; i++) {  
  
            for (int j = 1; j <= i; j++) {  
  
                System.out.print(number + " ");  
  
                number++;  
            }  
  
            System.out.println();  
        }  
    }  
}
```

Question 22: Write a program to print the following pattern:

Code:

```
public class HollowSquare {

    public static void main(String[] args) {

        for (int i = 1; i <= 6; i++) {

            for (int j = 1; j <= 6; j++) {

                if (i == 1 || i == 6 || j == 1 || j == 6) {

                    System.out.print("*");

                } else {

                    System.out.print(" ");

                }

            }

            System.out.println();

        }

    }

}
```

Question 23: Write a program to print the following pattern:

Code:

```
public class Diamond {

    public static void main(String[] args) {
```

```

for (int i = 1; i <= 6; i++) {
    for (int j = 1; j <= 6 - i; j++) {
        System.out.print(" ");
    }
    for (int j = 1; j <= (2 * i - 1); j++) {
        System.out.print("*");
    }

    System.out.println();
}

for (int i = 5; i >= 1; i--) {
    for (int j = 1; j <= 6 - i; j++) {
        System.out.print(" ");
    }
    for (int j = 1; j <= (2 * i - 1); j++) {
        System.out.print("*");
    }
    System.out.println();
}
}
}

```

Question 24: Reverse a String

Code:

```
import java.util.Scanner;
```

```

public class ReverseString{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a string: ");
        String str= sc.nextLine();
        String reverse="";
        for(int i=str.length()-1;i>=0;i--){
            reverse=reverse+str.charAt(i);
        }
        System.out.println("Reversed string: "+reverse);
    }
}

```

Question 25: Count Vowels in a String

Code:

```

import java.util.Scanner;

public class CountVowels{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a string: ");
        String str = sc.nextLine();
        int count=0;
        for (int i =0;i<str.length();i++){
            char ch = str.charAt(i);
            if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u'){
                count++;
            }
        }
    }
}

```

```

        }
    }
    System.out.println("The number of vowels in '"+str+"' is: "+count);
}
}

```

Question 26: Check if a String is a Palindrome

Code:

```

import java.util.Scanner;

public class Palindrome{

    public static void main(String args[]){

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a string: ");

        String str = sc.nextLine();

        str=str.toLowerCase();

        Boolean palindrome=true;

        for (int i=0;i<str.length();i++){

            if(str.charAt(i)!=str.charAt(str.length()-1-i)){

                palindrome=false;

            }

        }

        if(palindrome){

            System.out.println("The string '"+str+"' is palindrome.");

        } else{

            System.out.println("The string '"+str+"' is not a palindrome.");

        }

    }

}

```

```
    }  
}
```

Question 27: String Literal and Object Creation

Code:

```
import java.util.Scanner;  
  
public class StringLiteral {  
  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.println("Enter a string1: ");  
        String str1 = sc.nextLine();  
        System.out.println("Enter a string2: ");  
        String str2 = sc.nextLine();  
  
        System.out.println(  
            "Both variables point to the same object: " + (str1 == str2)  
        );  
    }  
}
```

Question 28: String Creation with new Keyword

Code:

```
public class StringCreation {
```

```
public static void main(String[] args) {  
  
    String str1 = new String("hello");  
    String str2 = new String("hello");  
  
    System.out.println("Using == : " + (str1 == str2));  
    System.out.println("Using .equals(): " + str1.equals(str2));  
}  
}
```

Question 29: String Concatenation and Object Creation

Code:

```
public class StringConcatenation {  
  
    public static void main(String[] args) {  
  
        String str1 = "hello";  
        String str2 = "world";  
  
        String str3 = str1 + str2;  
  
        System.out.println("Is str3 pointing to the same object as str1? "  
            + (str3 == str1));  
    }  
}
```

Question 30: String Pool with intern() Method

Code:

```
public class StringIntern {  
  
    public static void main(String[] args) {  
  
        String str1 = new String("hello");  
  
        String str2 = str1.intern();  
  
        String str3 = "hello";  
  
        System.out.println("Is str2 and str3 pointing to the same object? "  
            + (str2 == str3));  
    }  
}
```

Question 31: Multiple String Literals with Same Content

Code:

```
public class MultipleStringLiterals {  
  
    public static void main(String[] args) {  
  
        String str1 = "java";  
  
        String str2 = "java";  
  
    }  
}
```



```
String str3 = "java";
```

```
System.out.println("All strings point to the same object: "
```

```
    + (str1 == str2 && str2 == str3));
```

```
}
```

```
}
```